

Vishal Charran

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Education

CUNY - Queens College
Bachelor in computer science

Flushing, NY
Graduated May 2025

- **Relevant Course Work:** Data Structures, Software Engineering, Database Systems, Object-Oriented Programming in C++ & Java, Design and Analysis of Algorithms, Algorithmic Problem-Solving, Computer Organization and Assembly Language, Computer Architecture

Skills

- **Languages & Frameworks:** Python, Java, C++, Swift, Kotlin, JavaScript, Spring Boot, .NET, TypeScript, Node.js, React, React Native, SwiftUI, Next.js, Express.js, Flask, Django, JavaFX, HTML, CSS, XML, TensorFlow, PyTorch, Hugging Face, NumPy
- **Databases & Cloud:** SQL, MySQL, SQLite, MongoDB, Firebase, Milvus, AWS, Vercel
- **Development Tools:** Android Studio, Xcode, Selenium, Linux, GitHub, Docker, Fastlane, Figma, Jest, Cypress, JUnit, XCTest, Maven
- **Specialties:** Full-Stack Development, Web Development, Mobile Development, CI/CD, UI/UX Collaboration

Work Experience

Research Foundation (580 million net assets) | *Machine Learning Engineer*

Jul – Sep 2024 | Queens, NY

- Optimized **embedding**, chunking, and vector representations through rigorous testing and research, leading to **30%** improved IVF results.
- Engineered a custom **RAG** system with a vector database, improving AI response consistency from **80%** to **98%**.
- Led a team to develop a discord bot, leveraging **RAG** for **context-aware** results that outperformed **ChatGPT** in tasks like debugging code.
- Received recognition from **IBM** for developing a project with potential to become an enterprise-grade application on IBM's marketplace.

Queens College Tech Incubator | *Software Engineer Intern*

Oct 2022 - Jul 2023 | Queens, NY

- Led a cross-functional team of interns using Agile methodologies to build an AI-driven chatbot, increasing user engagement by **35%** and reducing support response time by **40%**.
- Enhanced chatbot search accuracy by implementing data tracking, enabling users to find relevant information **50% faster**, boosting customer satisfaction by **35%**.
- Executed unit testing and debugging procedures, increasing software efficiency by **30%** and ensuring seamless performance across devices.
- Earned a return offer for Fall 2023 due to exceptional technical contributions and leadership.

Code for All QC (Computer Science Club) | *Software Developer and Tutor*

Sep 2022 - Jul 2024 | Queens, NY

- Built and launched the club's official website serving **1,900+ members**, implementing React and Tailwind CSS for a modern, responsive design that increased engagement and event participation. (Website: codeforall.nyc)
- Increased student GPAs by **55%** (**3.7 vs. 2.4 class average**) through **one-on-one and group tutoring** in computer science.

Projects (All demos and GitHub repo available here: https://github.com/VishalRommel/Project_Demo)

AI Mediation Mobile App | (React Native/Expo, Python, Flask, MongoDB, Vector DB, OpenAI API)

- Built a cross-platform mobile app for AI-guided mediation, supporting private 1-to-1 and group sessions with contextual AI responses.
- Implemented secure **auth** & identity with **JWT**, profile endpoints, display-name resolution, contacts graph, and a 5-digit invite flow for seamless multi-device session joining.
- Designed and deployed chat data model & **REST APIs** (/session, /invite, /rooms, /contacts) with message persistence, private routing, **cached** display-names, incremental history, and robust networking (safe-JSON, retry/polling, **deep-linking**).

Artificial Intelligence Career Coach | (*Best AI project Cuny Hackathon 2024*) | (Python, Flask, React, Milvus)

- Devised a full-stack app with Python, Milvus, Flask, and React, enabling smooth communication between backend and frontend technologies.
- Spearheaded a team to engineer an AI tool leveraging **Milvus database** to analyze LinkedIn, GitHub, and resume data for tailored job-matching.
- Developed a dynamic resume generator by querying user data against job descriptions, ensuring highly relevant and personalized resumes.
- Implemented a vector search system to rank job compatibility by matching user resumes to real-time job postings, identifying the top opportunities from **1,000** recent listings in the user's field with the highest hiring potential.

Artificial Intelligence Exercise/Meal Prep Mobile App | (Java, Open AI API, SQL, Android Studio)

- Incorporated the OpenAI API to provide personalized plans to users, leveraging AI-driven recommendations for a tailored experience.
- Utilized AI to create a system that generates personalized weekly meals and exercise plans tailored to user-provided inputs.
- Modeled a database using MySQL to efficiently manage user profiles, preferences, and progress data.

Stock Trading Website - Real Time Stock Data | (JavaScript, Vantage API, Firebase)

- Engineered an interactive stock market simulator web application using JavaScript, HTML/CSS, and **Firebase**, allowing users to practice trading stocks with fake money on **1000s of stocks**.
- Integrated **Alpha Vantage REST API** to pull real-time stock price data for dynamic user portfolio tracking and performance analysis.
- Implemented user authentication with **Firebase** to securely store user account information and trading portfolio data in the cloud.